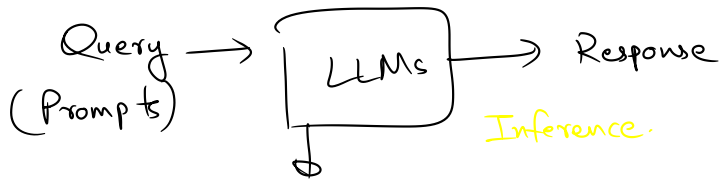
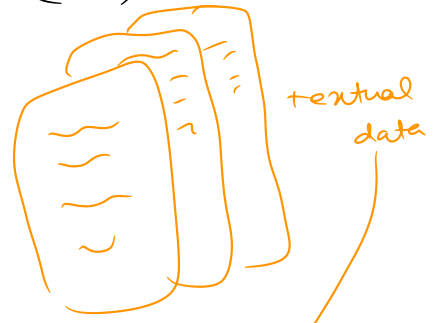


Retrieval Augmented Generation (RAG)



Large language Models.

Transformer based Neural Networks



LLMs: huge number of parameters.

weights & biases

Parametric knowledge

Wikipedia, GitHub, Litman-Internet

Challenges (Vanilla LLM):

pretrained on a large corpus of textual data

1) Prompt "How old is Harry?"
Harry Potter

"What are procedures that are followed in Amazon?"

2) knowledge cut off data

Pretraining LLM: GPT-4 GPT-3

Oct 2024
2025
13th Jan 2026

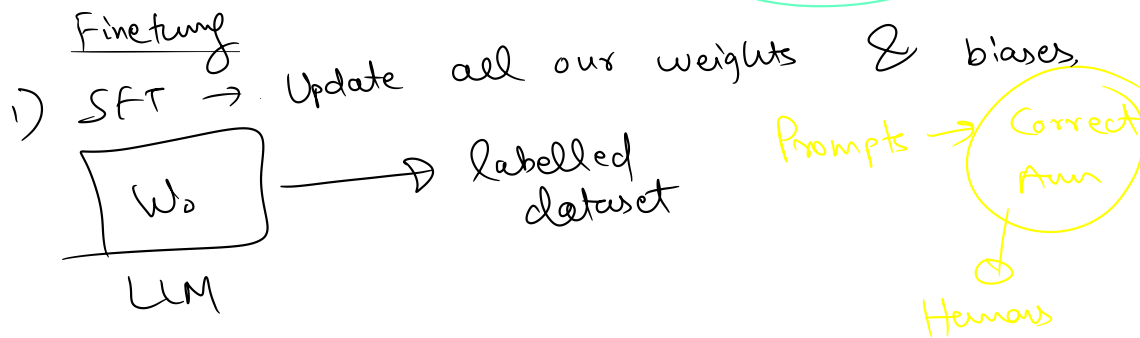
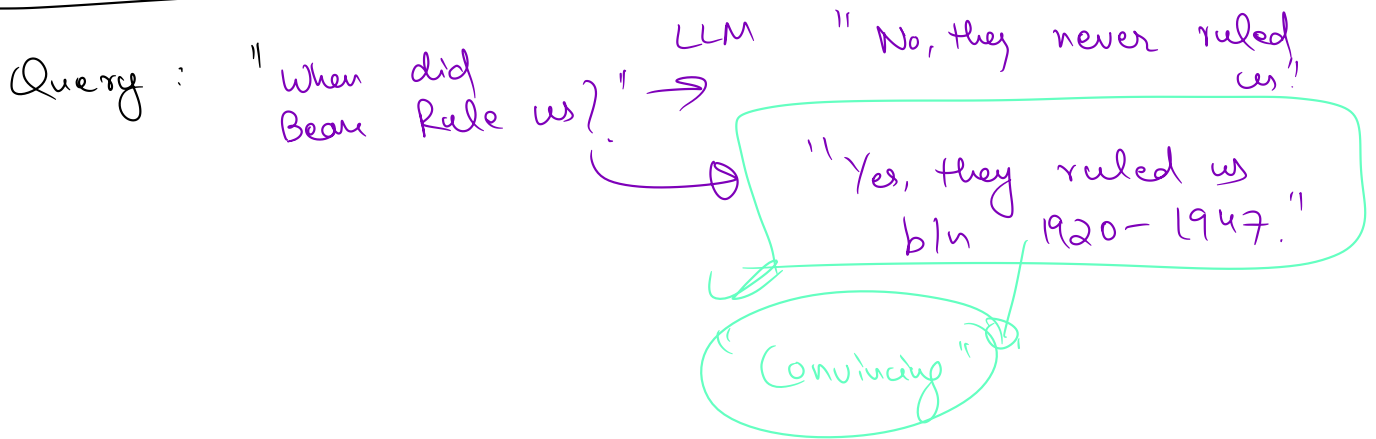
LLM has no access to the data.

"What is today's weather in Kanpur?"
14th Jan

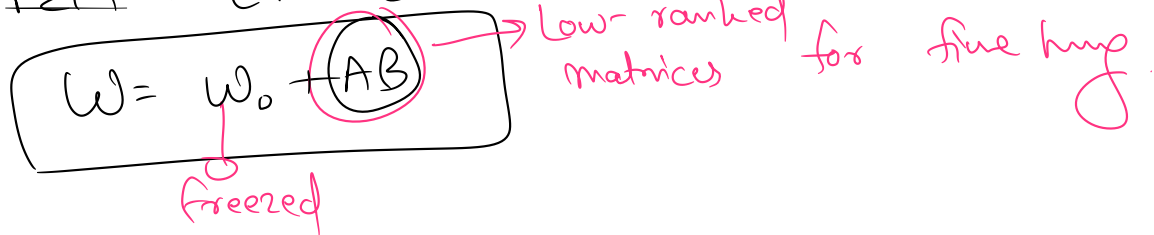
3) Hallucinations

Response probabilistic

convincing answer -



2) PEFT: LoRA & QLoRA



✓ 1) Private data: Amazon

Documents, Files, legal doc of Amazon

✓ 2) Knowledge Cut-off date:

Regularly update our LLM.

✓ 3) Hallucination:

India's in 100 BCE - 2026

"No, they never Rule"

Dynasty, Rule, &

Finetuning

Challenge

- SFT : we are updating all of our weights
- 1) Computational Cost: LORA & QLoRA: A & B
 - 2) Pretraining: English Literature. LLM French Novels.
(Fine tune) :
 - 3) Technical Expertise: From LLM → +ve sentiment.
Labeled data :

"Language Models are few Shot Learners"

Complex ↑ → emergent properties

IN CONTEXT LEARNING

It is the core capability of LLMs, where the model learns (GPT-3/4, Claude, Llama) to solve a task purely by seeing examples in the prompt — without updating the weights.

Eg

Query: Do sentiment Analysis on the following CSU file.

Examples

Please go through the below examples & learn how to judge Sentiments

- 1) I love my singing → Positive Sentiment
- 2) I hate my college → Negative Sentiment
- 3) My life sucks → Negative

CSU contains: "I enjoy fishing" → ??

GPT-1
GPT-2

(Basic)

GPT-3

Few Shot Pm
One Shot Prompting

Emergent Property: it is a behaviour or ability that suddenly appears in a system when it reaches a certain scale of complexity, even though it was not explicitly programmed on that task.

Context

Query

Query

External Databases

Prompt

LLM

Response =

Query + Relevant

Context

14 videos

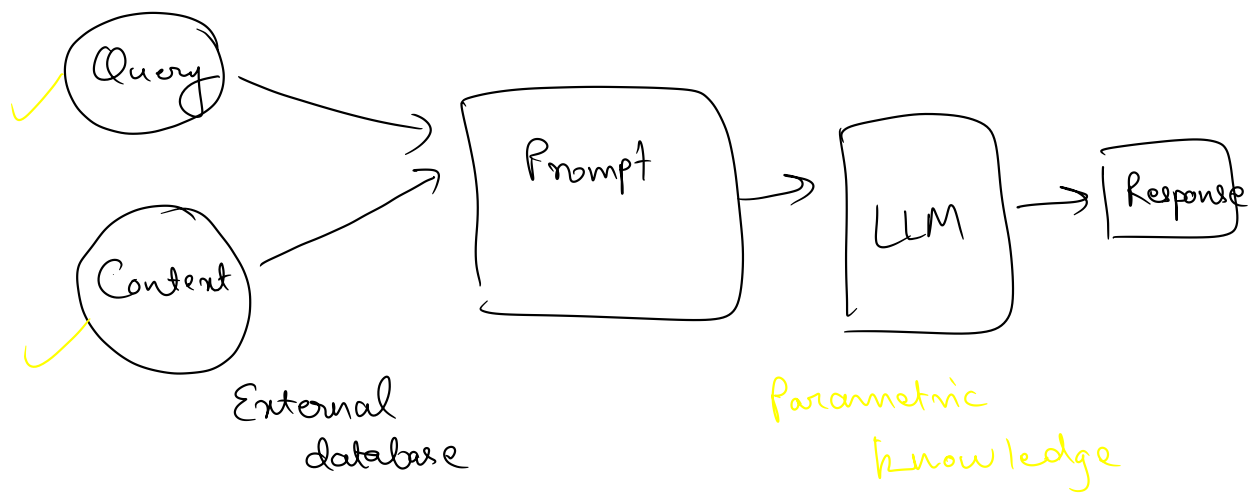
1 video

→ Response

Youtube "My Tensors"
What is Bahdanau Attention?

~~_____~~
~~_____~~
~~_____~~
~~_____~~

RAg (Retrieval Augmented Generation)



RAg: Retrieval + Generation

Indexing: It is the process of preparing our knowledge base so that it can be efficiently searched up, at the time of query

Architected Intelligence

Course Materials } All the Assignments.
↓
PDFs, PPTs }

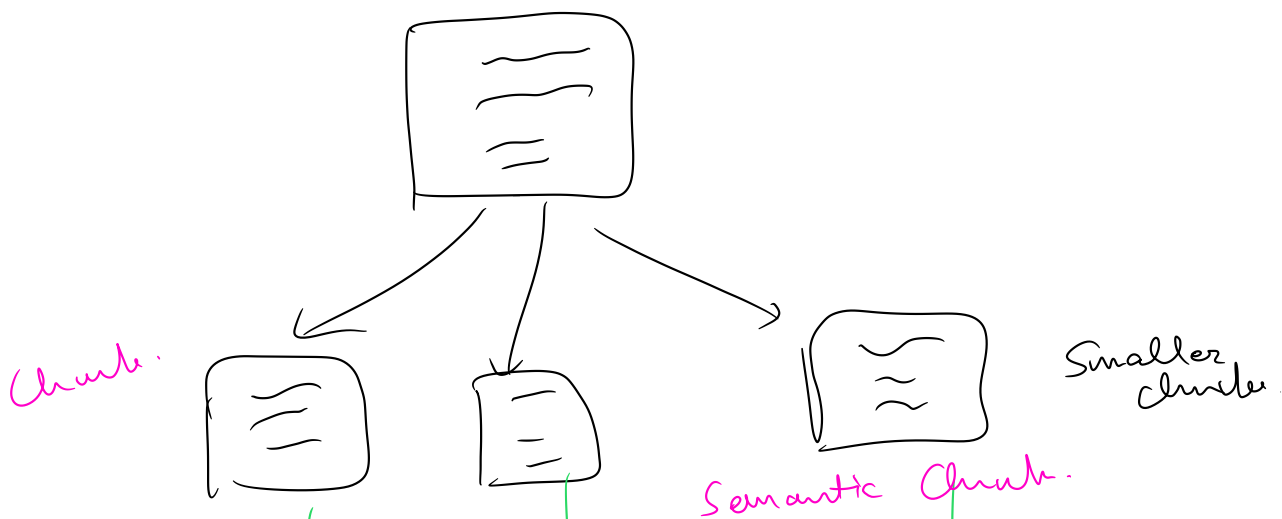
4 sub-steps

1) Document Ingestion:
You load your source knowledge into the memory.

PDF documents
Youtube Transcripts
Github Repos.

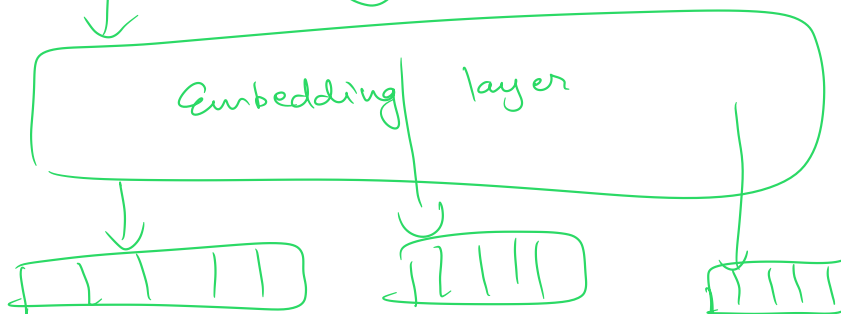
LangChain: PyPDF Loader, GitLoader, etc. =

2) Text chunking: Break our large documents into small, semantically meaning chunks.



3) Embedding generation: Convert each chunk into a dense vector that captures its semantic meaning

512 dim



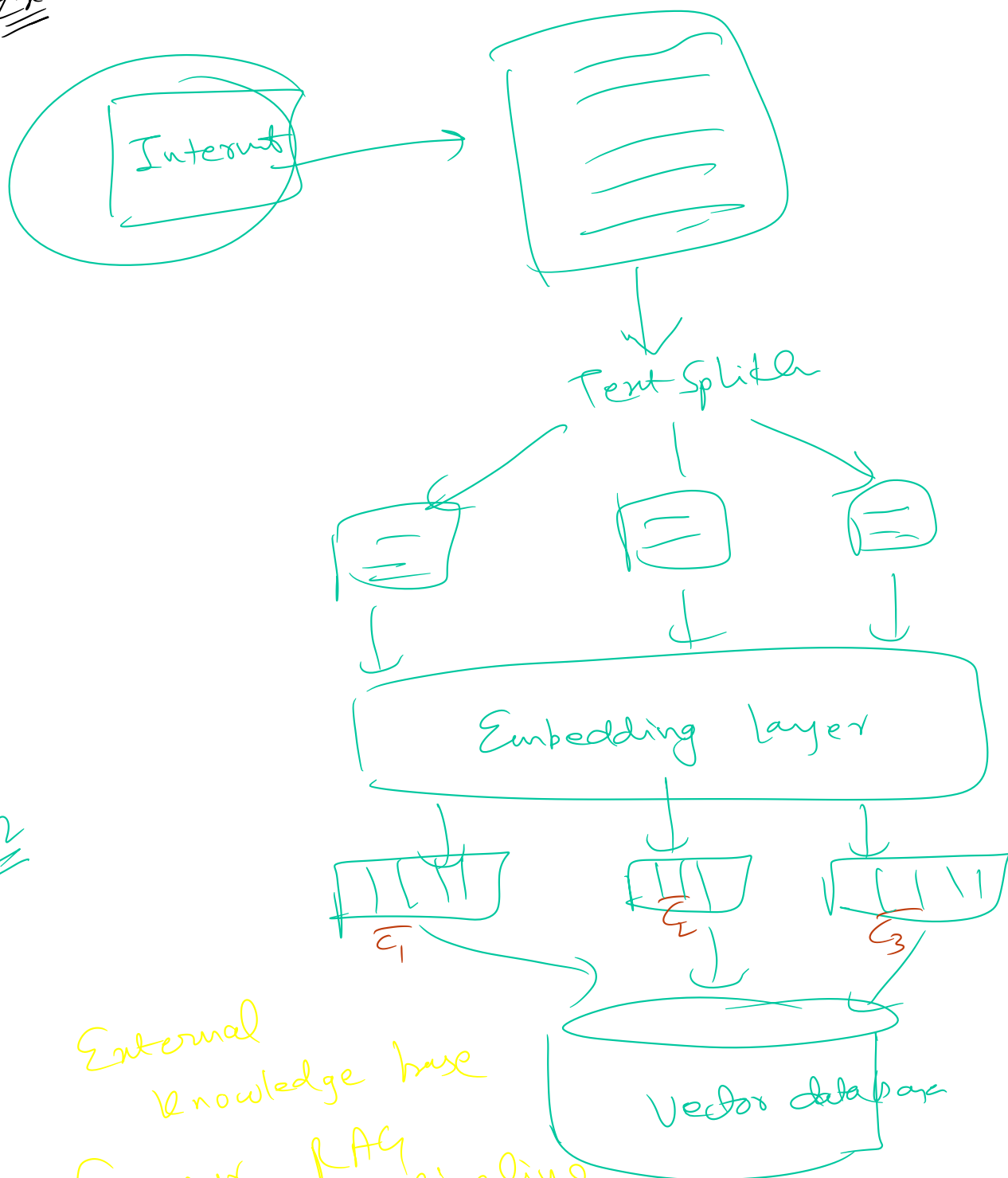
Ex OpenAI Embedding
Gemini GenAI

4) Store : Original chunk + vector
embedding

Vector database.

Chroma, FAISS, etc.

Ex



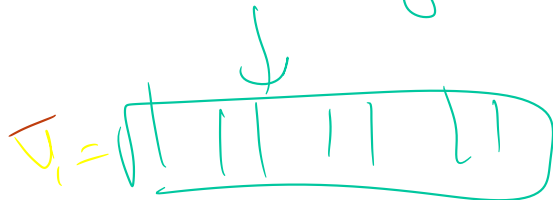
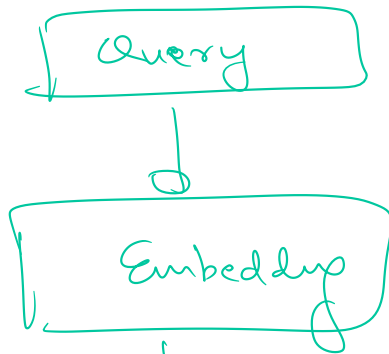
5/2

2

Retrieval:

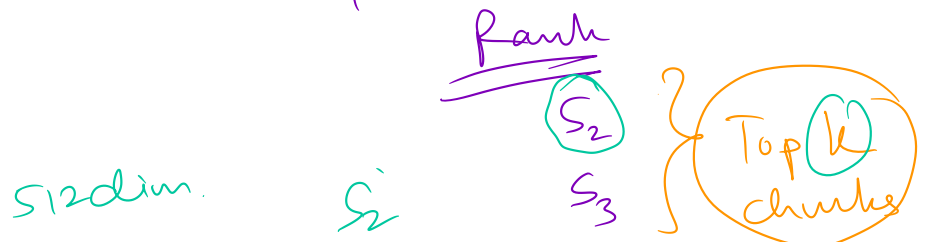
finding the most relevant pieces of information from a pre-built index

Cosine Similarity, Euclidean distance, BM25, vector database



$$\frac{\overline{V_1} \cdot \overline{C_1}}{\|V_1\| \|C_1\|}, \frac{\overline{V_1} \cdot \overline{C_2}}{\|V_2\| \|C_2\|}, \frac{\overline{V_1} \cdot \overline{C_3}}{\|V_3\| \|C_3\|}$$

$S_1 \quad S_2 \quad S_3$



Step-1: embedding of our query vector

Step-2: Semantic Search

Step-3: Ranking (Top-k)

Step-4: Most relevant

3

Augmentation

Where the retrieval documents (chunks of relevant content) are combined with the user's query to form a new, enriched prompt for the LM.

Example

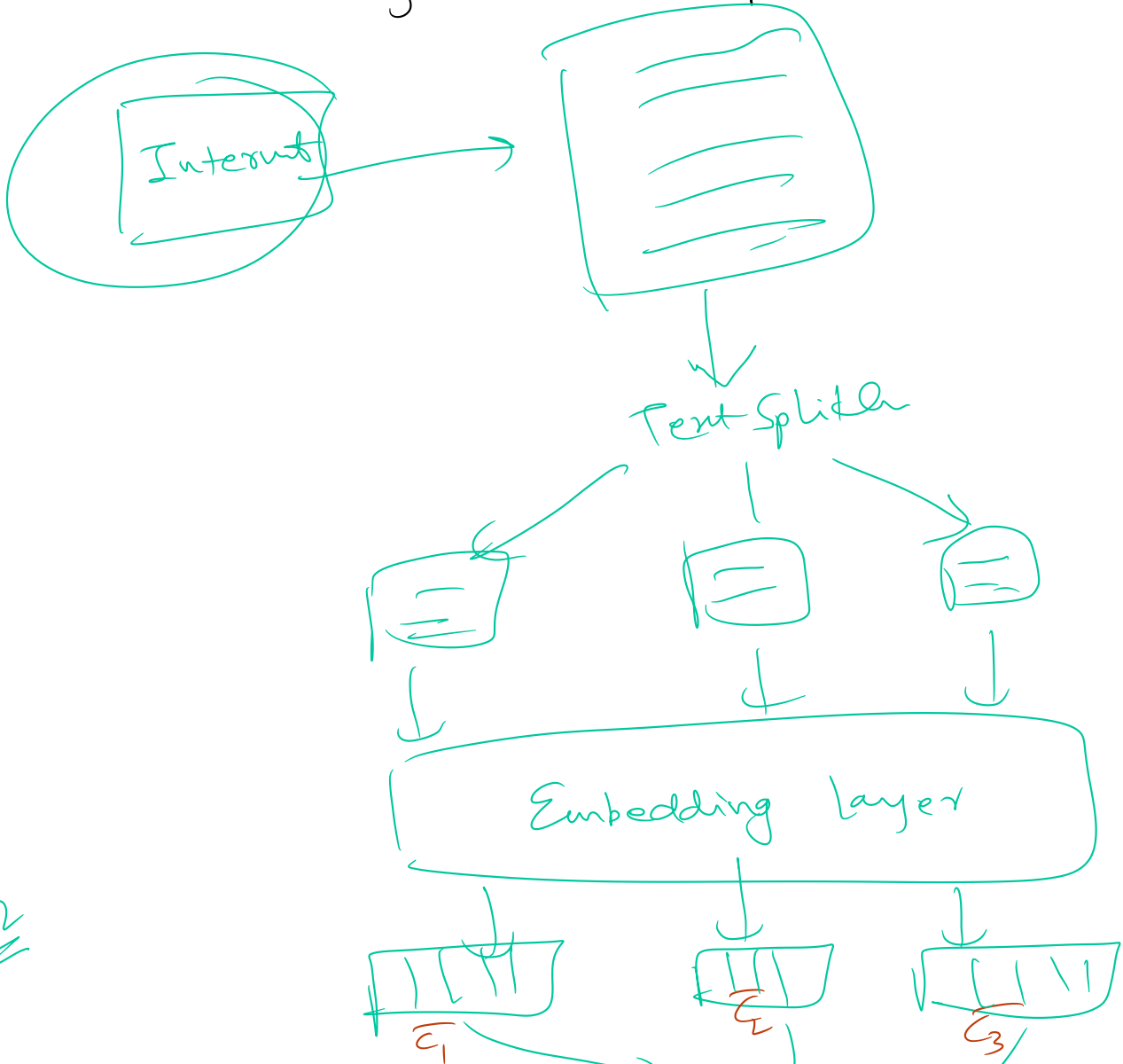
" " " You are an helpful assistant

Answer the question ONLY from the provided context. If the context is insufficient, just say you don't know.

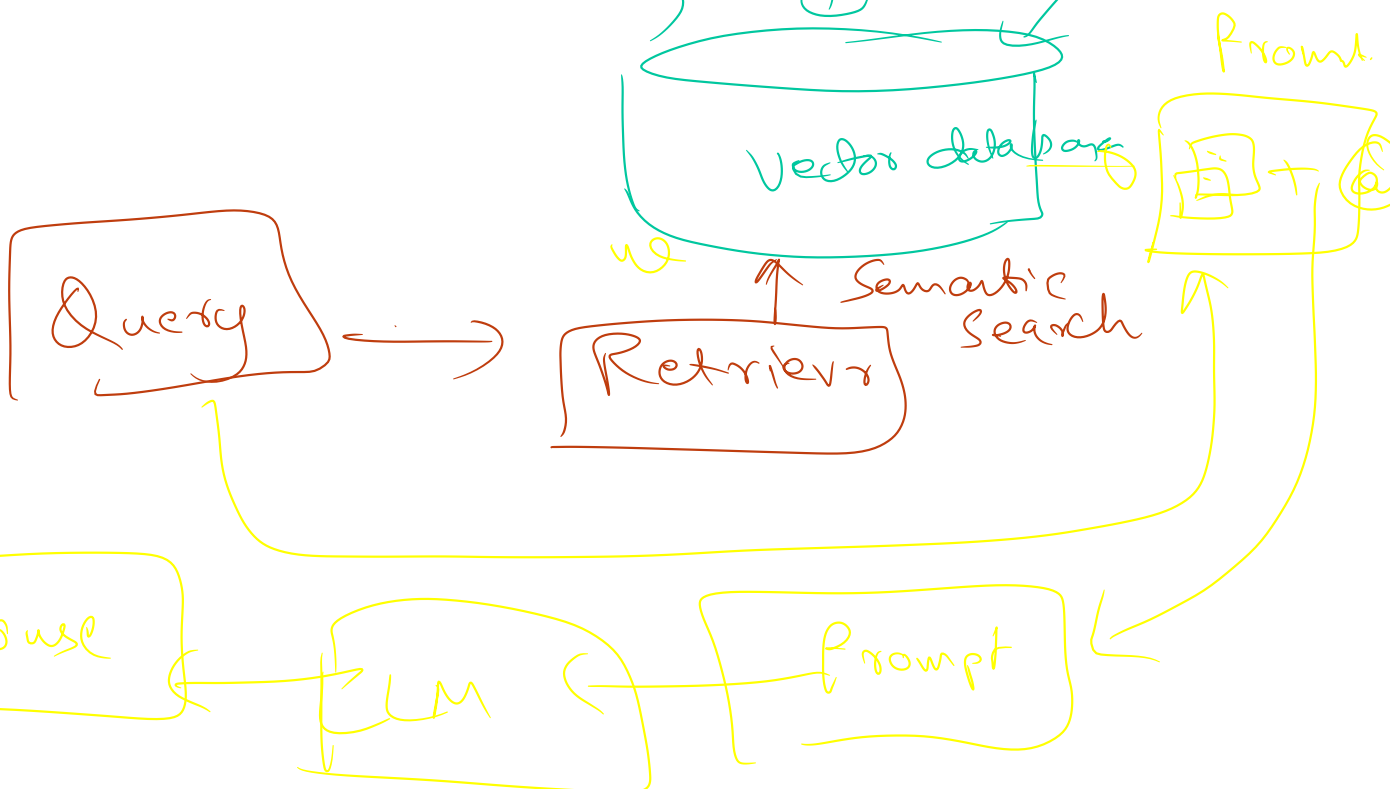
{ context }

Question, { question } " " "

④ Generation LLM uses the user's query & retrieved & augmented context to generate a response



5/12



Vanilla LLM RAG

1) Private data



2) Knowledge Cut off date

LLM: 1st Jan 2026

14th Jan 2026

Weather

3) Hallucinations

"Teddy Bear"

No they never ruled us

India's history



Knowledge base

Legal

Medical field

App^h
RAG